Ritik Parihar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.98
Shanti Niketan Public School	Class XII (CBSE)	2023	90.80%
Shanti Niketan Public School	Class X (CBSE)	2021	90.00%

TECHNICAL SKILLS

Programming Languages: Python, C++, SQL

Machine Learning: Supervised Learning, Unsupervised Learning, Linear Regression, Logistic Regression, Decision Trees, Random Forest, Gradient Boosting, Feature Engineering, Model Evaluation

Deep Learning: Neural Networks, Backpropagation, CNNs, Attention Mechanism, Transformer Architecture

AI/ML Libraries & Frameworks: NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow

Data Visualization: Matplotlib

Data Structures & Algorithms: Arrays, Strings, Hashing, Binary Search, Stacks, Queues, Linked Lists, Trees, Graphs, Dynamic Programming, Graph Algorithms

Core Computer Science: Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

Developer Tools: Git, GitHub, VS Code, Jupyter Notebook, Google Colab

PROJECTS

Explainable Corporate Financial Distress Early-Warning System | ML, Data Science, Finance

- Developed an **end-to-end financial-risk screening system** integrating data preprocessing, financial-ratio **feature engineering**, model comparison, explainability, dashboard development, and deployment.
- Designed a **decision-threshold framework** based on review-capacity constraints rather than the default 0.5 threshold, balancing recall with the proportion of borrowers flagged for review.
- Evaluated models using **business-focused performance metrics** and documented target-label definitions, model limitations, and non-causal interpretation of predictions to support transparent model usage.

Book Recommendation System | Python, Pandas, Scikit-learn, Streamlit

- Developed a **hybrid recommendation system** combining popularity-based and collaborative-filtering approaches to generate personalized book recommendations from user-rating data.
- Built the recommendation pipeline covering **data preprocessing, feature preparation, recommendation generation, and user interaction** using Pandas, NumPy, and Scikit-learn.
- Designed and deployed an **interactive Streamlit application** on Streamlit Cloud, integrating the complete workflow from user input to recommendation output and managing the project through GitHub.

Smart Delivery Route Optimizer -- **Python | Graph Algorithms | Route Optimization**

- Built a **route optimization application** using weighted graphs to model delivery locations and road connections, supporting route generation and distance calculation across multiple delivery points.
- Implemented **Dijkstra's shortest-path algorithm** to compute minimum-distance routes and developed reusable modules for graph construction, route processing, and distance calculation.
- Developed **pytest-based automated tests** covering shortest-path computation, route generation, and edge cases, with a modular codebase designed for maintainability and future extensions.

CODING PROFILES

LeetCode: 400+ problems solved | **Profile**

EXPERIENCE

Private Tutor | Summer Vacation

- Taught and mentored **25-30 students** through structured classes, simplifying complex concepts and adapting explanations to students' learning needs.
- Planned lessons, conducted practice sessions, and monitored student progress while developing **communication, leadership, and time-management skills**.

RELEVANT COURSES

Computer Science: Introduction to Programming, Data Structures and Applications, Computational Physics

AI/ML: Introduction to Artificial Intelligence , Machine Learnig for Physics , Machine Learning , Deep Learning

Mathematics: Probability, Statistics, Linear Algebra, Calculus

Electronics: Digital Electronics, Basic Electrical Engineering ,Systems and Signal Processing

Other: Communication Skills, Introduction to Entrepreneurship, Macroeconomics

EXTRACURRICULAR

NSS Volunteer: Participated in NSS activities and contributed to community-oriented initiatives.

Fest Operations: Assisted in organizing and coordinating **cultural and technical college fests**, supporting event operations and execution.

Sports: Represented the **hostel and department in inter-college volleyball tournaments**.

HOBBIES AND INTERESTS

Sports & Reading: Playing volleyball and reading fiction and non-fiction.

Cinema & Finance: Watching cinema and following financial markets.